



MATHEMATICS (EXTENSION 1)

Preliminary Course Assessment Task 2

April 4, 2012

General instructions

- Working time – 1 hour and 15 minutes.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 9)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NAME: # BOOKLETS USED:

Class (please ✓)

☐ 11M3A – Mr Lin/Mr Rezcallah

☐ 11M3D – Mr Lucas

☐ 11M3B – Mr Fletcher

☐ 11M3E – Mr Lam

☐ 11M3C – Ms Ziazaris

☐ 11M3F – Mr Berry

Marker's use only.

QUESTION(S)	1-10	11	12	13	14	15	16	Total	%
MARKS	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{91}$	

Section I: Objective response

Mark your answers on the multiple choice sheet (numbered as page 9) provided. **Marks**

1. If $y = |x + 4|$, then $y = 0$ for **1**

(A) $x = -4$

(C) $x = 4$

(B) $x = 0$

(D) no real values of x

2. If $\sin \theta = \frac{1}{2}$ and $90^\circ \leq \theta \leq 180^\circ$, then $\cos \theta =$ **1**

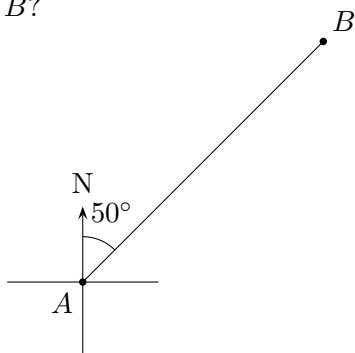
(A) $\frac{\sqrt{3}}{2}$

(B) $-\frac{\sqrt{3}}{2}$

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

3. The bearing of Town B from Town A is 050° . What is the bearing of Town A from Town B ? **1**



(A) 050°

(B) 040°

(C) 220°

(D) 230°

4. $(x + 1)(x - 3)(x - 5) > 0$ if **1**

(A) $-1 < x < 3$ only

(C) $-1 < x < 3$ or $x > 5$

(B) $x < -1$ or $3 < x < 5$

(D) $3 < x < 5$ only

5. The linear function $f(x) = 6 - 2x$ has range $R = \{y : -4 \leq y \leq 12\}$. **1**

The domain is thus

(A) $D = \{x : x \in \mathbb{R}\}$

(C) $D = \{x : -3 \leq x \leq 5\}$

(B) $D = \{x : -5 \leq x \leq 3\}$

(D) none of the above

6. $f(x - 2) + 4$ shifts the original curve of $f(x)$ **1**

(A) $\leftarrow 2$ units and $\downarrow 4$ units

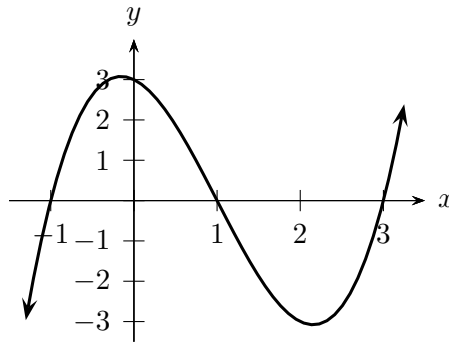
(C) $\rightarrow 2$ units and $\downarrow 4$ units

(B) $\leftarrow 2$ units and $\uparrow 4$ units

(D) $\rightarrow 2$ units and $\uparrow 4$ units

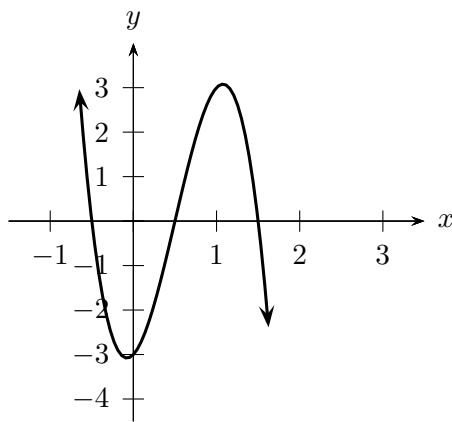
7. The graph of $y = f(x)$ is shown. The graph which best represents $1 - f(2x)$ is

1

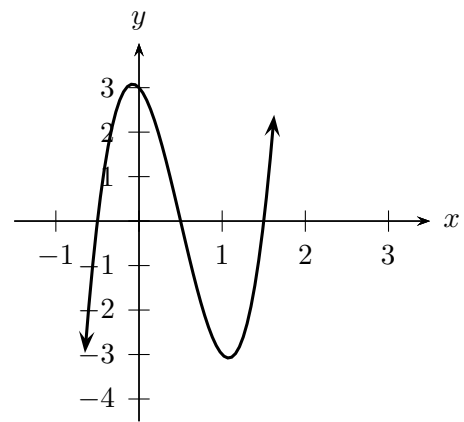


(A)

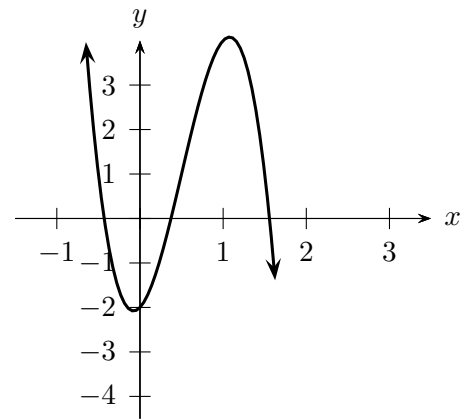
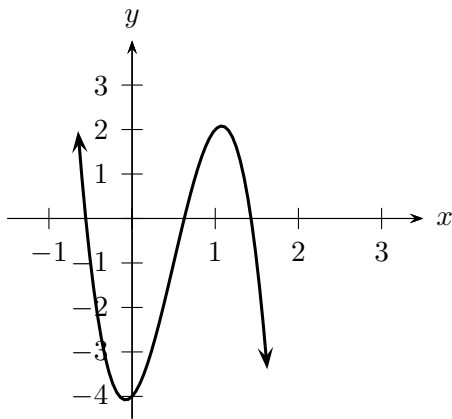
(C)



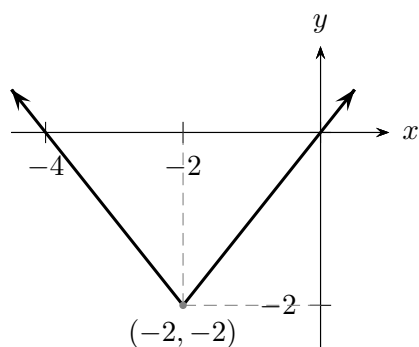
(B)



(D)



8. The rule that defines the function shown in the diagram is 1



(A) $y = |x - 2| - 2$

(B) $y = |x - 2| + 2$

(C) $y = |x + 2| - 2$

(D) $y = |x + 2| + 2$

9. The equation of the vertical asymptotes to $f(x) = \frac{x^2 - 4}{x^2 + 2x - 3}$ are 1

(A) $x = -1, x = 3$

(C) $x = -1, x = -3$

(B) $x = 1, x = -3$

(D) $x = 1, x = 3$

10. A circle with centre $(a, -2)$ and radius 5 has the equation 1

$$x^2 - 6x + y^2 + 4y = b$$

where a and $b \in \mathbb{R}$. The value of a and b are respectively:

(A) -3 and 38

(C) -3 and -8

(B) 3 and 12

(D) -3 and 0

Examination continues overleaf...

Section II: Short answer

Question 11 (15 Marks)

Commence a NEW page.

Marks

- (a) Simplify $|3| - |-4| + |3 - 4|$ **1**
- (b) For the formula $a = \sqrt{\frac{2V^2 - 1}{y}}$, find a correct to 4 significant figures given $V = 640$ and $y = 9.65$. **2**
- (c) Solve $7x^2 - 4x - 1 = 0$, giving your answer in exact form. **2**
- (d) Express $0.1\overline{234}$ as a fraction in simplest form. **3**
- (e) If $p = 38.6 \times 10^{-5}$ and $q = 471.3 \times 10^{-8}$, find the value of pq in scientific notation correct to 2 decimal places. **2**
- (f) Factorise $8t^3 - 1\,000$ fully. **2**
- (g) Solve $\left| \frac{x+1}{3} - \frac{x+2}{4} \right| = 5$ **3**

Question 12 (15 Marks)

Commence a NEW page.

Marks

- (a) Factorise fully:
- i. $4a^2 - 9(b+2)^2$ **3**
 - ii. $t^3 + t^2 - t - 1$ **4**
- (b) Solve for x :
- i. $x^2 < 3x$ **2**
 - ii. $|2x - 3| > 1$ **3**
 - iii. $8^{2x-1} = 16^{x+1} \times 4^{-x}$ **3**

Question 13 (15 Marks)	Commence a NEW page.	Marks
(a) Expand $(5 - 2\sqrt{3})^2$.		2
(b) Rewrite in simplest form, with a rational denominator:		2
$\frac{1}{2\sqrt{3}-1} + \frac{3}{\sqrt{3}+1}$		
(c) Simplify $\frac{p^2+2p}{3p^2+5p-2} \div \frac{9p+3}{9p^2-1}$ fully.		3
(d) Solve for a and b if		4
$a + b + 2\sqrt{3} = 4 + \sqrt{2a - 2b}$		
(e) Solve the inequation $\frac{1}{1-x} \leq \frac{1}{1+x}$.		4

Question 14 (12 Marks)	Commence a NEW page.	Marks
(a) Write the exact value of		
i. $\cos 120^\circ$.		2
ii. $\operatorname{cosec} 675^\circ$		3
(b) Solve for x : $\sin 4x = \cos(6x + 50)$.		2
(c) Show that $\tan x - \tan x \sin^2 x = \sin x \cos x$.		2
(d) Solve for θ , where $0^\circ \leq \theta \leq 360^\circ$:		3
$2 \cos^2 \theta + 3 \cos \theta - 2 = 0$		

Question 15 (12 Marks)

Commence a NEW page.

Marks

- (a) In $\triangle PQR$, $PQ = 20$ cm, $QR = 22$ cm and $\angle PRQ = 15^\circ$. Find the value(s) for $\angle RPQ$ and PR . **4**
- (b) $ABCD$ is a parallelogram with $AB = 8$ cm, $BC = 5$ cm and $\angle DAB = 60^\circ$. Calculate the exact area of the parallelogram. **3**
- (c) A passenger on a small cruise ship observed, whilst passing under the Harbour Bridge, that the angle of elevation to the pylons are 47° and 28° . It is known that the distance between the pylons is 503 m. (Assume the pylons are equal in height and elevation.)



- i. By drawing an appropriate diagram, labelling sides and assigning variables, show that the height h of the pylons is given by **3**

$$h = \frac{503 \tan 47^\circ \tan 28^\circ}{\tan 47^\circ + \tan 28^\circ}$$

- ii. Hence or otherwise, find the height of the pylon in centimetres, using scientific notation correct to 4 significant figures. **2**

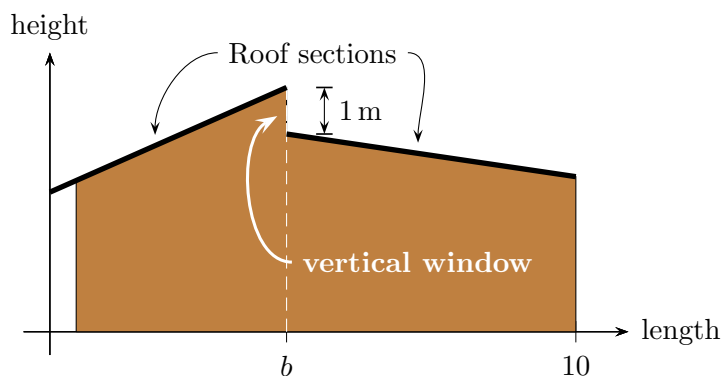
Examination continues overleaf...

Question 16 (12 Marks) Commence a NEW page. **Marks**

- (a) Expand and fully simplify **2**

$$(a - 5)^2 + (a - 4)^2 + \cdots + a^2 + \cdots + (a + 4)^2 + (a + 5)^2$$

- (b) Two roof sections of a building are separated by a vertical window as shown in the diagram. **3**



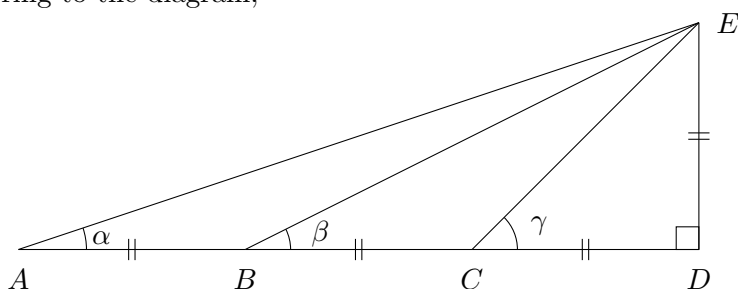
The roof sections are modelled by

$$\text{height} = \begin{cases} \frac{1}{2} \times \text{length} + 3 & 0 \leq \text{length} < b \\ -\frac{1}{6} \times \text{length} + 5 & b \leq \text{length} \leq 10 \end{cases}$$

The vertical window has height 1 m. Find the value of b .

- (c) For the function $f(x) = |x^2 - 9| + 3$
- State the domain and range. **2**
 - Sketch the function. **2**

- (d) By referring to the diagram,



- Find the value of γ . **1**
- By using $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$, show that $\alpha + \beta = \gamma$. **2**

🐼 End of paper. 🐼

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

NAME:

Class (please ✓)

☐ 11M3A – Mr Lin/Mr Rezcallah

☐ 11M3D – Mr Lucas

☐ 11M3B – Mr Fletcher

☐ 11M3E – Mr Lam

☐ 11M3C – Ms Ziazaris

☐ 11M3F – Mr Berry

- 1 – ☐ A ☐ B ☐ C ☐ D
- 2 – ☐ A ☐ B ☐ C ☐ D
- 3 – ☐ A ☐ B ☐ C ☐ D
- 4 – ☐ A ☐ B ☐ C ☐ D
- 5 – ☐ A ☐ B ☐ C ☐ D
- 6 – ☐ A ☐ B ☐ C ☐ D
- 7 – ☐ A ☐ B ☐ C ☐ D
- 8 – ☐ A ☐ B ☐ C ☐ D
- 9 – ☐ A ☐ B ☐ C ☐ D
- 10 – ☐ A ☐ B ☐ C ☐ D

Suggested Solutions

Section I

1. (A) 2. (B) 3. (D) 4. (C) 5. (C)
6. (D) 7. (D) 8. (C) 9. (B) 10. (B)

Section II

Question 11 (Ziaziaris)

(a) (1 mark)

$$|3| - |-4| + |3 - 4| = 3 - 4 + 1 = 0$$

(b) (2 marks)

- ✓ [1] for magnitude.
- ✓ [1] for 4 significant figures.

$$a = 291.4$$

(c) (2 marks)

- ✓ [1] for substituting into quadratic formula.
- ✓ [1] for final answer.

$$\begin{aligned} 7x^2 - 4x - 1 &= 0 \\ x &= \frac{-(-4) \pm \sqrt{4^2 - 4(7)(-1)}}{2 \times 7} \\ &= \frac{4 \pm \sqrt{16 + 28}}{14} \\ &= \frac{4 \pm \sqrt{44}}{14} \\ &= \frac{2 \pm \sqrt{11}}{7} \end{aligned}$$

(d) (3 marks)

- ✓ [1] for multiplying by correct power of 10.
- ✓ [1] for subtraction
- ✓ [1] for final answer.

$$\text{Let } \underset{\times 1\,000}{x} = \underset{\times 1\,000}{0.1\overline{234}} \quad (1)$$

$$1000x = 123.4\overline{234} \quad (2)$$

(2) - (1)

$$\begin{aligned} \frac{999x}{\div 999} &= \frac{123.3}{\div 999} \\ \therefore x &= \frac{123.3}{999} = \frac{1\,233}{9\,990} = \frac{137}{1\,110} \end{aligned}$$

(e) (2 marks)

- ✓ [1] for magnitude.
- ✓ [1] for correct scientific notation.

$$(38.6 \times 10^{-5}) \times (471.3 \times 10^{-8}) = 1.82 \times 10^{-9}$$

(f) (2 marks)

- ✓ [-1] for each error (including not being fully factorised)

$$\begin{aligned} 8t^3 - 1\,000 &= 8(t^3 - 125) \\ &= 8(t - 5)(t^2 + 5t + 25) \end{aligned}$$

(g) (3 marks)

- ✓ [1] for simplifying fraction inside abs val.
- ✓ [1] for each correct solution.

$$\begin{aligned} \left| \frac{x+1}{3} - \frac{x+2}{4} \right| &= 5 \\ \left| \frac{4x+4-3x-6}{12} \right| &= 5 \\ \left| \frac{x-2}{12} \right| &= \frac{5}{\times 12} \\ \therefore |x-2| &= 60 \\ x-2 &= \pm 60 \\ \therefore x &= 2 \pm 60 = -58 \text{ or } 62 \end{aligned}$$

Question 12 (Lucas)

(a) i. (3 marks)

- ✓ [-1] for each error, including not fully simplified to the last step.

$$\begin{aligned} 4a^2 - 9(b+2)^2 &= (2a - 3(b+2))(2a + 3(b+2)) \\ &= (2a - 3b - 6)(2a + 3b + 6) \end{aligned}$$

ii. (4 marks)

- ✓ [-1] for each error, including not fully factorised + simplified to the last step.

Question 13 (Fletcher)

(a) (2 marks)

- ✓ [-1] for each error.

$$\begin{aligned} t^3 + t^2 - t - 1 &= t^2(t+1) - (t+1) \\ &= (t+1)(t^2-1) \\ &= (t+1)(t-1)(t+1) \\ &= (t+1)^2(t-1) \end{aligned}$$

(b) (2 marks)

- ✓ [-1] for each error.

$$\begin{aligned} (5 - 2\sqrt{3})^2 &= 25 - 20\sqrt{3} + 12 \\ &= 37 - 20\sqrt{3} \end{aligned}$$

(b) i. (2 marks)

- ✓ [1] for $x(x-3) < 0$
 ✓ [1] for final answer.
 ✓ [0] if student cancels one of the x terms.

$$\begin{aligned} x^2 &< 3x \\ x^2 - 3x &< 0 \\ x(x-3) &< 0 \\ \therefore 0 &< x < 3 \end{aligned}$$

$$\begin{aligned} &\frac{1}{2\sqrt{3}-1} \times \frac{2\sqrt{3}+1}{2\sqrt{3}+1} + \frac{3}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} \\ &= \frac{2\sqrt{3}+1}{12-1} + \frac{3\sqrt{3}-3}{3-1} \\ &= \frac{2\sqrt{3}+1}{11} \times \frac{2}{2} + \frac{3\sqrt{3}-3}{2} \times \frac{11}{11} \\ &= \frac{4\sqrt{3}+2+33\sqrt{3}-33}{22} \\ &= \frac{37\sqrt{3}-31}{22} \end{aligned}$$

ii. (3 marks)

- ✓ [1] for correct cases.
 ✓ [1] for each correct solution.

$$\begin{aligned} |2x-3| &> 1 \\ 2x-3 &> 1 & \quad 2x-3 < -1 \\ 2x &> 4 & \quad 2x < 2 \\ x &> 2 & \quad x < 1 \\ \therefore x &> 2 \text{ or } x < 1 \end{aligned}$$

(c) (3 marks)

- ✓ [1] for factorising.
 ✓ [1] for cancelling.
 ✓ [1] for final answer.

$$\begin{aligned} &\frac{p^2+2p}{3p^2+5p-2} \div \frac{9p+3}{9p^2-1} \\ &= \frac{\cancel{p(p+2)}}{\cancel{(3p-1)(p+2)}} \times \frac{\cancel{(3p-1)}(3p+1)}{\cancel{3(3p+1)}} \\ &= \frac{p}{3} \end{aligned}$$

iii. (3 marks)

- ✓ [1] for converting to base 2
 ✓ [1] for correct expansion of indices
 ✓ [1] for final answer.

$$\begin{aligned} 8^{2x-1} &= 16^{x+1} \times 4^{-x} \\ (2^3)^{2x-1} &= (2^4)^{x+1} \times (2^2)^{-x} \\ 2^{6x-3} &= 2^{4x+4} \times 2^{-2x} = 2^{2x+4} \\ 6x-3 &= 2x+4 \\ 4x &= 7 \\ \therefore x &= \frac{7}{4} \end{aligned}$$

(d) (4 marks)

- ✓ [1] for equating rational & irrational parts.
 ✓ [1] for setting up simultaneous equations.
 ✓ [1] for each of a and b .

$$a+b+2\sqrt{3}=4+\sqrt{2a-2b}$$

Rational parts:

$$\begin{aligned} a+b &= 4 \\ 2a+2b &= 8 \end{aligned} \quad (13.1)$$

Irrational parts:

$$\begin{aligned}\sqrt{2a-2b} &= 2\sqrt{3} = \sqrt{12} \\ \therefore 2a-2b &= 12 \quad (13.2)\end{aligned}$$

$$(13.2) - (13.1)$$

$$\begin{aligned}-4b &= 4 \\ \therefore b &= -1 \\ \therefore a &= 5\end{aligned}$$

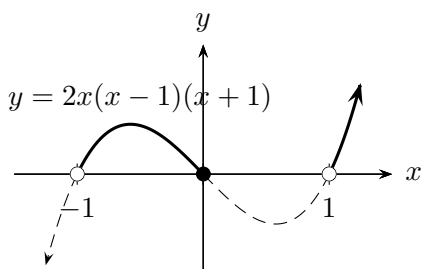
(e) (4 marks)

- ✓ [1] for multiplying by square of denominator on each side.
- ✓ [2] for all critical points. [-1] for each error.
- ✓ [1] for correct $<$, $\leq$ and $>$ sign.

$$\frac{1}{1-x} \leq \frac{1}{1+x} \\ \times (1-x)^2(1+x)^2 \quad \times (1-x)^2(1+x)^2$$

Note $x \neq \pm 1$

$$\begin{aligned}(1-x)(1+x)^2 &\leq (1+x)(1-x)^2 \\ (1+x)(1-x)^2 - (1-x)(1+x)^2 &\geq 0 \\ (1+x)(1-x)((1-x) - (1+x)) &\geq 0 \\ -2x(1+x)(1-x) &\geq 0 \\ 2x(x-1)(x+1) &\geq 0\end{aligned}$$



$$\therefore -1 < x \leq 0 \text{ or } x > 1$$

Question 14 (Lam)

(a) i. (2 marks)

- ✓ [1] for correct sign.
- ✓ [1] for correct magnitude.

$$\cos 120^\circ = -\frac{1}{2}$$

ii. (3 marks)

- ✓ [1] for subtracting 360°
- ✓ [1] for correct sign.
- ✓ [1] for correct magnitude.

$$\operatorname{cosec} 675^\circ = \frac{1}{\sin 315^\circ} = \frac{1}{-\frac{1}{\sqrt{2}}} = -\sqrt{2}$$

(b) (2 marks)

- ✓ [1] for realising the angles are complementary.
- ✓ [1] for final answer.

$$\sin 4x = \cos (6x + 50)$$

$$4x + (6x + 50) = 90$$

$$10x + 50 = 90$$

$$10x = 40$$

$$x = 4$$

(c) (2 marks)

- ✓ [-1] for each error.

$$\begin{aligned}\tan x - \tan x \sin^2 x \\ &= \tan x (1 - \sin^2 x) \\ &= \frac{\sin x}{\cancel{\cos x}} \times \cancel{\cos x} \\ &= \sin x \cos x\end{aligned}$$

(d) (3 marks)

- ✓ [1] for factorising correctly.
- ✓ [1] for each correct solution.

$$2 \cos^2 \theta + 3 \cos \theta - 2 = 0$$

Let $m = \cos \theta$,

$$2m^2 + 3m - 2 = 0$$

$$(2m-1)(m+2) = 0$$

$$2m = 1 \quad m = -2$$

$$\therefore 2 \cos \theta = 1 \quad \cos \theta = -2$$

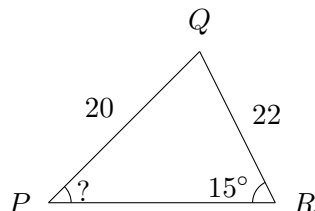
As $-1 \leq \cos \theta \leq 1$, second solution is invalid.

$$\begin{aligned}\therefore \cos \theta &= \frac{1}{2} \\ \theta &= 60^\circ, 300^\circ\end{aligned}$$

Question 15 (Berry)

✓ [1] for doubling area of triangle.

(a) (4 marks)

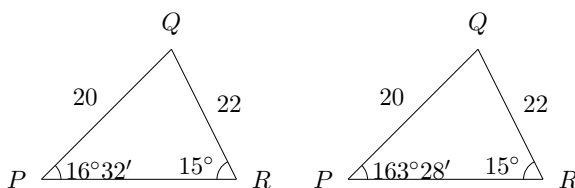
✓ [1] for each correct value of $\angle RPQ$ and PR .

$$\frac{\sin \angle RPQ}{22} = \frac{\sin 15^\circ}{20}$$

$$\sin \angle RPQ = \frac{22 \sin 15^\circ}{20}$$

$$\angle RPQ = 16^\circ 32' \text{ or } 163^\circ 28'$$

(there are two solutions as the angle to be found is opposite the longer side)



(Not to scale)

Using the angle sum of $\triangle$, $\angle PQR = 148^\circ 28'$ or $1^\circ 32'$.

- When $\angle PQR = 148^\circ 28'$,

$$\frac{PR}{\sin 148^\circ 28'} = \frac{20}{\sin 15^\circ}$$

$$PR = \frac{20 \sin 148^\circ 28'}{\sin 15^\circ}$$

$$= 40.42 \dots$$

- When $\angle PQR = 1^\circ 32'$,

$$\frac{PR}{\sin 1^\circ 32'} = \frac{20}{\sin 15^\circ}$$

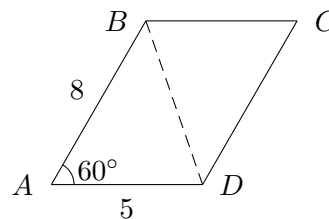
$$PR = \frac{20 \sin 1^\circ 32'}{\sin 15^\circ}$$

$$= 2.08 \dots$$

(b) (3 marks)

✓ [1] for correct area of triangle via sine formula.

✓ [1] for area of triangle as exact value.

Area of $\triangle ABD$:

$$A_{\triangle ABD} = \frac{1}{2} \times 5 \times 8 \times \sin 60^\circ$$

$$= 10\sqrt{3}$$

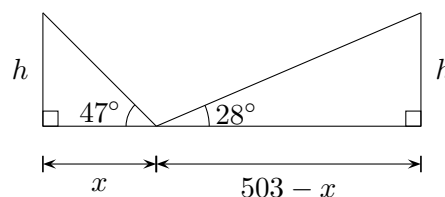
$$\therefore A_{ABCD} = 20\sqrt{3}$$

i. (3 marks)

✓ [1] for correct diagram

✓ [1] for two equations in x , and $503 - x$

✓ [1] for final answer.

Let x be the distance from the souther pylon to the ship (could be from the northern pylon, but this is not important as the solution is identical)

$$\frac{h}{x} = \tan 47^\circ$$

$$\therefore x = \frac{h}{\tan 47^\circ} \quad (15.1)$$

In the ther triangle,

$$\frac{h}{503 - x} = \tan 28^\circ$$

$$\therefore 503 - x = \frac{h}{\tan 28^\circ}$$

$$x = 503 - \frac{h}{\tan 28^\circ} \quad (15.2)$$

Equating x in (15.1) and (15.2)

$$\begin{aligned} 503 - \frac{h}{\tan 28^\circ} &= \frac{h}{\tan 47^\circ} \\ 503 &= \frac{h}{\tan 47^\circ} + \frac{h}{\tan 28^\circ} \\ &= h \left(\frac{1}{\tan 47^\circ} + \frac{1}{\tan 28^\circ} \right) \\ &= h \left(\frac{\tan 28^\circ + \tan 47^\circ}{\tan 28^\circ \tan 47^\circ} \right) \end{aligned}$$

Dividing both sides by $\frac{\tan 28^\circ + \tan 47^\circ}{\tan 28^\circ \tan 47^\circ}$,

$$\therefore h = \frac{503 \tan 28^\circ \tan 47^\circ}{\tan 28^\circ + \tan 47^\circ}$$

ii. (2 marks)

- ✓ [1] for correct number of significant figures.
- ✓ [1] for correct scientific notation.
- ✓ [-1] for metres (required to give cm)

$$h = 178.79 \text{ m} = 1.788 \times 10^4 \text{ cm}$$

Question 16 (Rezcallah)

(a) (2 marks)

- ✓ [1] expanding some terms correctly *or* obtaining one term of the answer correct.
- ✓ [1] correct final answer ([1] only for entire question if no working is provided)

$$(a-5)^2 = a^2 - 10a + 25$$

$$(a+5)^2 = a^2 + 10a + 25$$

$$(a-4)^2 = a^2 - 8a + 16$$

$$(a+4)^2 = a^2 + 8a + 16$$

⋮

Observations

- there will be eleven a^2 terms (including the single $(a+0)^2$ expansion)
- all a terms are offset;
- Constant term produces double of sum of squares $-2(1^2 + 2^2 + \dots + 5^2)$

$$\begin{aligned} \therefore (a-5)^2 + \dots + a^2 + \dots + (a+5)^2 \\ = 11a^2 + 110 \end{aligned}$$

(b) (3 marks)

- ✓ [2] for correct equation $\frac{1}{2}b + 3 = (-\frac{1}{6}b + 5) + 1$.
- ✓ [1] for correct final answer.

- Upper part of the roof, at b :

$$h = \frac{1}{2}b + 3$$

- Lower part of the roof, at b :

$$h = -\frac{1}{6}b + 5$$

- Relating the upper and lower part (these parts differ by 1 m):

$$\frac{1}{2}b + 3 = \left(-\frac{1}{6}b + 5 \right) + 1$$

$$\underbrace{\frac{1}{2}b + 3}_{\times 6} = \underbrace{-\frac{1}{6}b + 6}_{\times 6}$$

$$3b + 18 = -b + 36$$

$$4b = 18$$

$$\therefore b = \frac{9}{2}$$

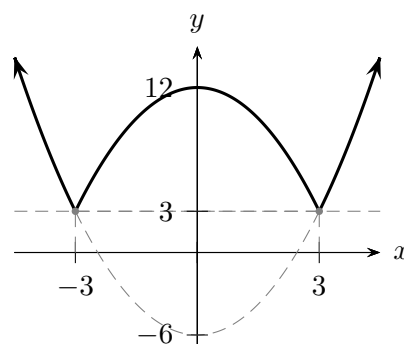
(c) i. (2 marks)

$$D = \{x : x \in \mathbb{R}\}$$

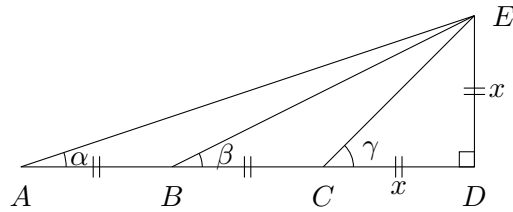
$$R = \{y : y \geq 3\}$$

ii. (2 marks)

- ✓ [1] for correct shape of the curve.
- ✓ [1] for correct intercepts and critical points with the shape.



(d) i. (1 mark)



$$\tan \gamma = \frac{x}{x} = 1$$

$$\therefore \gamma = 45^\circ$$

ii. (2 marks)

✓ [1] for both $\tan \alpha = \frac{1}{3}$ and $\tan \beta = \frac{1}{2}$ and substituting into given formula.

✓ [1] for showing $\tan(\alpha + \beta) = 1 = \tan \gamma$ using the given formula.

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \left(\frac{1}{2} \times \frac{1}{3}\right)} \\ &= \frac{\frac{5}{6}}{1 - \frac{1}{6}} \\ &= 1 \end{aligned}$$

$$\therefore \alpha + \beta = 45^\circ$$

$$\therefore \alpha + \beta = \gamma$$